ECHO — Case Report

Entangled Minds: The Quantathon · Qubits Qrew, NED University
Investigator: Inam (GitHub: Inam72) **Team:** [TEAM NAME]
Modules completed: 01, 03, 05 (mandatory) · 04 (stretch) · 02 Bronze (extra)

1. Final Theory / Verdict

ECHO never sent a message faster than light. What Dr. Zara Mahmud's twin detectors showed was real quantum entanglement: two outcomes that always agree. Agreement is *correlation*, not *communication*. Each detector on its own produces pure 50/50 noise, and neither side can choose its result, so no message can be encoded in it. To even *see* the pattern, the two sides must compare results over an ordinary channel, which travels at or below light speed.

Human side: our best explanation is that Zara **mistook correlation for a link**. Her evidence ("blue for blue, red for red, never a mismatch") is exactly what a Bell pair produces, and also exactly what a pre-agreed secret produces (Module 03). Bilal's missing report of a "discrepancy" fits this: someone had noticed the flaw. We do **not** have enough evidence to name a saboteur.

Confidence: *High (~90%)* that ECHO could not signal faster than light; this rests on our measurements plus the no-signalling theorem. *Moderate (~60%)* that the failure was self-deception rather than deliberate rigging, because the clue that would settle it (the seed circuit) was unavailable to us.

2. Evidence and Clues

- **Module 03:** an entangled pair and a classically pre-agreed pair both give only `00`/`11`. "Always agreeing", Zara's core evidence, cannot by itself prove a quantum link, let alone a message.
- **Module 03 Gold:** only the entangled pair still agrees after a change of measurement basis. The correlation is genuinely quantum, but each observer alone still sees random results.
- **Module 05:** the corrupted 96-bit stream decodes (majority vote) to **"ECHO"**, the name of her experiment, recovered intact.
- **Module 04:** measuring a superposition irreversibly changes it. A log that "changed when it was observed" is ordinary physics and is **not**, by itself, proof that investigators (or Sana) tampered with it. This weakens ARIA's accusation.
- **Module 05 Gold:** a qubit cannot be copied (CNOT entangles instead of copying), so no one could have "cloned" the signal to read it at both ends.

3. Evidence Board

FACT (observed)	CLAIM (stated by someone)	INFERENCE (our interpretation)	OPEN QUESTION
Bell pair gives only `00`/`11` (522 / 502 of 1024)	Zara: "entanglement *can* carry a message"	Her detectors showed correlation, not a signal	Who configured ARIA's run of ECHO?
Pre-agreed pair also gives only `00`/`11` (495 / 529)	Vance: "ECHO was never real"	Vance is right about the physics; that alone doesn't make him guilty	Was the "random" seed biased (Module 02 Silver)?
In the X basis: entangled pair still agrees (488 / 536); pre-agreed pair gives all four outcomes (~25% each)	Sana: "I only deleted a corrupted log"	"Changed when observed" is consistent with measurement collapse	What did the deleted log contain, and when was it deleted?
Decoded message = **ECHO**	Bilal: "I warned her" (report now missing)	Bilal likely spotted the correlation ≠ communication flaw	Where is Bilal's report?
Measuring mid-circuit turns a recoverable state into 50/50	ARIA: "No unauthorised command was issued"	ARIA's statement can be true *and* the run still misconfigured	Machine time of the final run vs. the log times

4. Experiments and Results

Module	Prediction	Circuit	Result
01 Bronze	X → always 1; H → 50/50	`x(0)`, `h(1)`, measure	`01`: 543, `11`: 481. Qubit 0 always 1
01 Silver	H·H → always 0	`h`, `h`, measure	`0`: 1024
01 Gold	`101`	`x([0,2])`, measure	`101`: 1024
03 Bronze	only `00`/`11`	`h(0)`, `cx(0,1)`	`00`: 522, `11`: 502
03 Silver	also only `00`/`11`	`h(0)`, **measure**, `cx(0,1)`	`00`: 495, `11`: 529
03 Gold	entangled agrees in X basis; secret doesn't	add `h([0,1])` before measuring	488/536 vs. 257/241/258/268
05 Bronze–Silver	majority vote recovers text; 2 flips break it	Python majority vote	"ECHO"; `100`→`110` corrupts the first letter
05 Gold	CNOT cannot copy + or −	`h`, `cx`, `h`, measure	copy +: `00`/`11` (513/511); copy −: `01`/`10` (518/506). Real copies would give only `00` / only `11`
04 Bronze–Silver	X·H·H → 1; H·X·H → 0; measure in the middle → 50/50	single-qubit H, X, measure	as predicted; mid-circuit measurement gives 50/50

All runs used the Qiskit Aer simulator, 1024 shots.

5. Interpretation

Module 03 is the crux. Its Silver result shows Zara's evidence could not distinguish a quantum link from a shared secret. Its Gold result shows her pair *was* genuinely entangled, but entanglement still does not let one side control what the other sees: each detector alone gives 50/50 regardless of what the other side does. The pattern only appears when the two result lists are brought together, which needs a classical channel. This is the **no-signalling theorem**: correlation without communication. Module 05 Gold adds that the state could not have been copied to "read" it at both ends, and Module 04 shows that the "changing" log is expected physics rather than evidence of a cover-up.

6. Alternative Explanation

Competing theory: deliberate rigging. Someone with access (Vance, who wanted ECHO exposed, or whoever configured ARIA) biased the "random" seed that triggered ECHO, so the run produced results that *looked* like a message.

Why we prefer self-deception: it explains everything we observed using only standard physics and Zara's own words ("…Or did I?"), and needs no extra actor. The rigging theory needs evidence we don't have, such as a measured bias in the seed circuit. Rigging also could not have made ECHO *truly* faster than light; at most it could have made it *look* successful.

7. Next Experiment

Run the seed-bias test (Module 02 Gold) on the original seed circuit. Measure P(1) and compute $z = (P(1) - 0.5) / \sqrt{(0.25/N)}$. A bias b needs about $2.25 / b^2$ shots to detect at 3σ ($b = 0.05$ needs ~900 shots). A significant bias supports rigging; a fair result supports self-deception. We would also run **Module 06** (teleportation with and without the classical bits) to show directly that the receiver's qubit is pure noise until the classical message arrives.

8. Limitations

- Our Module 03 Gold test uses only two bases (Z and X). A cleverer cheat that pre-agreed answers for *both* bases could pass it. Ruling out every such model needs the full Bell/CHSH test with intermediate measurement angles, which uses rotation gates not available in this module.
- All results come from a simulator, not real hardware.
- We did not have the organizers' preloaded seed circuit or the clues it unlocks (machine time, erased log contents, Zara's note), so we cannot identify or exclude any individual suspect.

9. References and AI/External Assistance

1. Qubits Qrew (2026). *ECHO: The Signal That Was Never Sent — Participant Manual.* NED University.
2. Nielsen MA and Chuang IL (2010). *Quantum Computation and Quantum Information*, 10th anniversary ed. Cambridge University Press.
3. Wootters WK and Zurek WH (1982). A single quantum cannot be cloned. *Nature* 299:802–803.
4. Bell JS (1964). On the Einstein Podolsky Rosen paradox. *Physics* 1:195–200.
5. Clauser JF, Horne MA, Shimony A and Holt RA (1969). Proposed experiment to test local hidden-variable theories. *Physical Review Letters* 23:880–884.
6. Qiskit documentation (2026). https://docs.quantum.ibm.com

AI assistance: Claude (Anthropic) was used to explain concepts, check reasoning and code, and help structure and draft this report. All circuits were run and checked by the investigator.